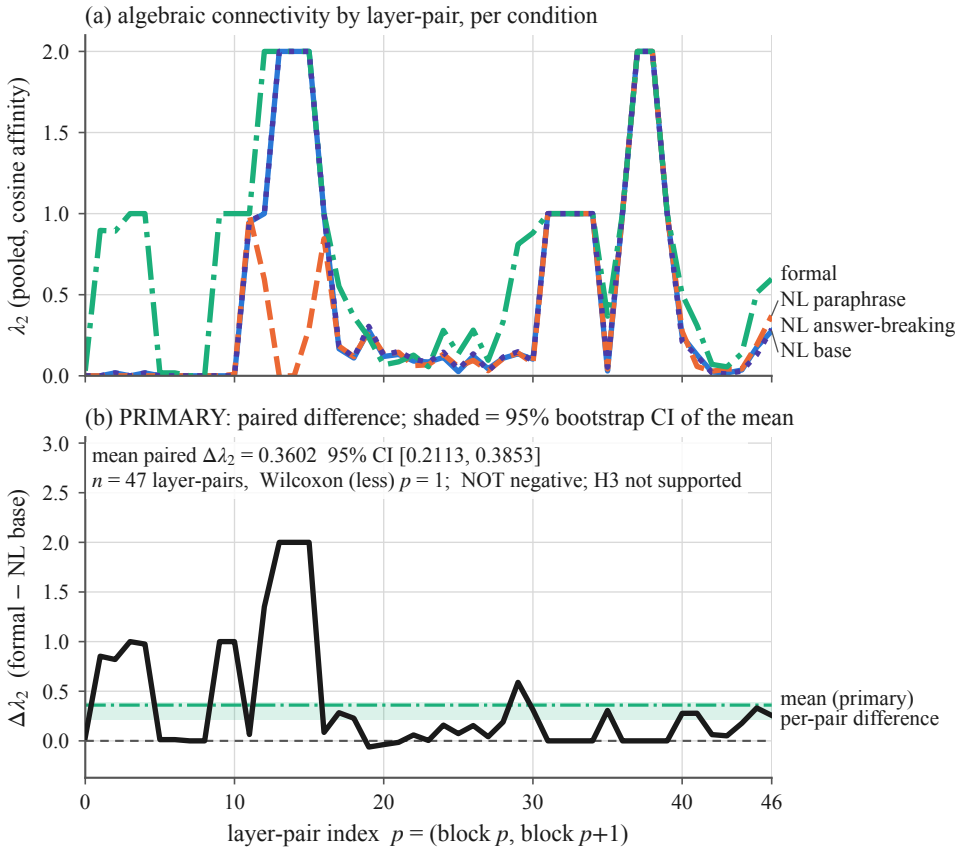
(a) algebraic connectivity by layer-pair, per condition
λ_2 (pooled, cosine affinity)
2.0
1.5
1.0
0.5
0.0
formal
NL paraphrase
NL answer-breaking
NL base
(b) PRIMARY: paired difference; shaded = 95% bootstrap CI of the mean
mean paired $\Delta\lambda_2$ = 0.3602 95% CI [0.2113, 0.3853]
n = 47 layer-pairs, Wilcoxon (less) p = 1; NOT negative; H3 not supported
$\Delta\lambda_2$ (formal − NL base)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
mean (primary)
per-pair difference
0
10
20
30
40
46
layer-pair index p = (block p, block p+1)